PCEOT402
ELECTRICAL MACHINES

Module 2

Transformers



Single Phase Transformer

- Constructional details
- Principle of operation
- EMF equation
- Ideal and practical transformer
- No-load and on-load operation
- Phasor diagrams
- Equivalent circuit
- Voltage regulation

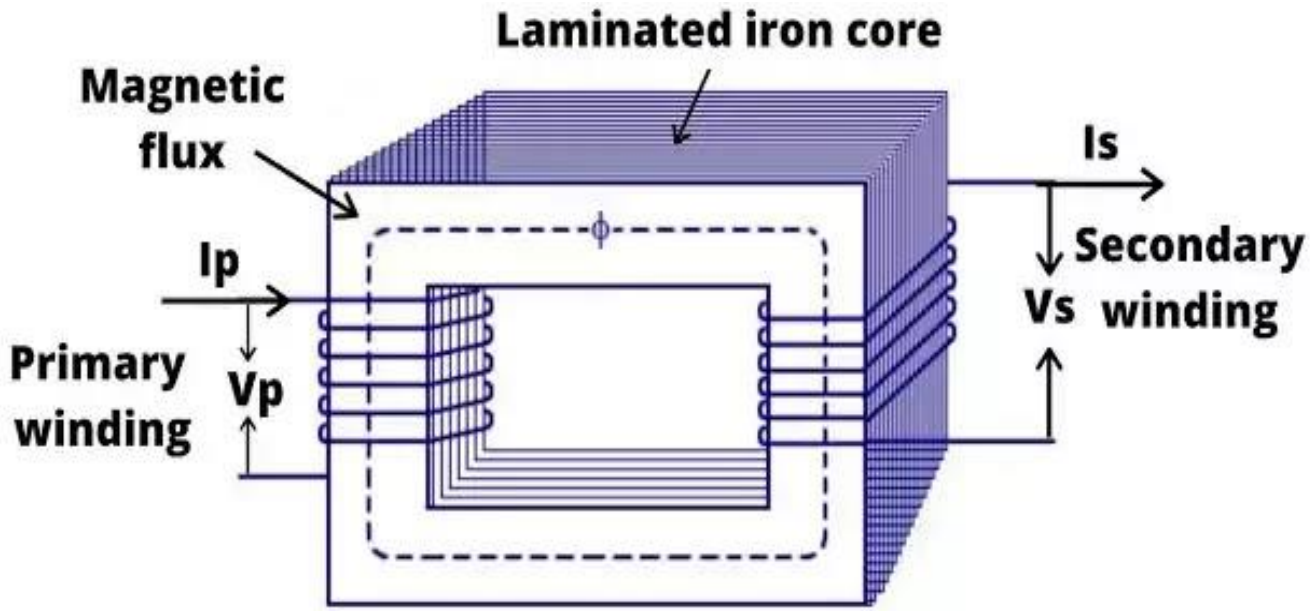
Principle

- Transformer – static device
- Electric power in one circuit is transformed into electric power of **same frequency** in another circuit
- principle of Faraday's Law of Electromagnetic Induction.
- Basic principle – Mutual Induction between 2 coils linked by a common flux
- The coils are **not** connected electrically but **linked** magnetically
- *Primary* – coil which is connected to the alternating voltage source
- *Secondary* – coil which is connected to the load

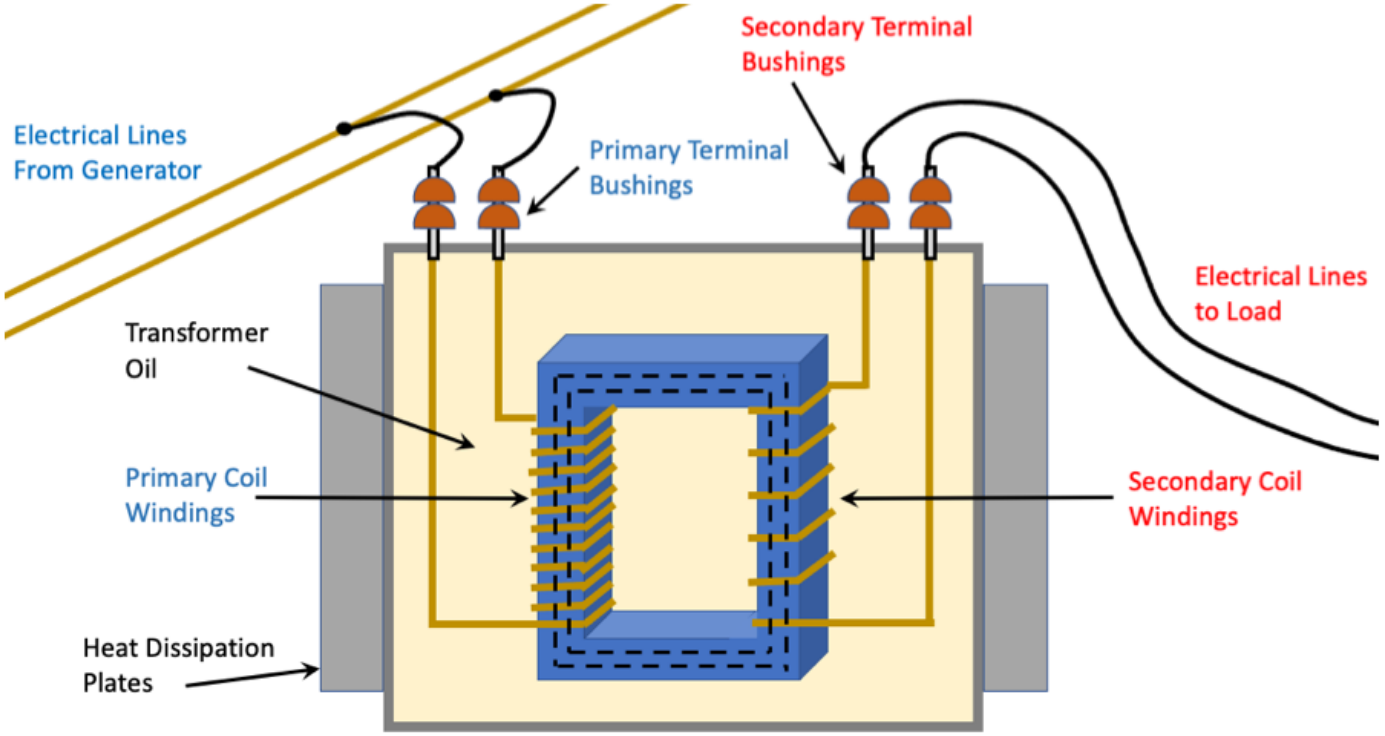
Principle

- Does not involve any rotating parts; so efficiency will be high
- For low capacity transformers, the efficiency is nearly 90-96% and for high capacity transformers, the efficiency is nearly 96-99%

Principle


Laminated iron core
Magnetic flux
Is
Ip
Secondary winding
Vs
Primary winding
Vp

Principle


Secondary Terminal Bushings
Electrical Lines From Generator
Primary Terminal Bushings
Electrical Lines to Load
Transformer Oil
Primary Coil Windings
Secondary Coil Windings
Heat Dissipation Plates

Principle

Transformer is a device which :-

- Transfers electric power from one circuit to another
- It transfers power without changing the frequency
- Achieves this with the help of electromagnetic induction
- The 2 electric circuits(primary & secondary) are in mutual inductive influence of each other

Types

1. based on voltage levels

- Step-up transformer
  - secondary voltage is greater than the primary voltage
  - number of turns of secondary winding is greater than the number of turns of primary winding
  - LV winding is the primary
- Step-down transformer.
  - secondary voltage is lesser than the primary voltage
  - number of turns of secondary winding is smaller than the number of turns of primary winding
  - HV winding is the primary
- Same transformer can be used step-up and step-down transformer

Types

2. Transformer based on core construction

- Core type transformer
- Shell type transformer

3. Transformer based on winding arrangement

- Autotransformer

  - single winding – shared by both the primary and the secondary sides

  - used to get a variable AC voltage

Types

4. Transformers based on usage.

- Distribution transformer

  - have ratings below 200 kVA

  - used to step down voltages at the distribution level

  -put in operation in the entire day irrespective of whether they carry load or not

  - iron losses will be present through out the day

  - copper losses will be present only when it is loaded

  - flux density is 1.7 Wb/m^2

  - voltage regulation is around 4-8%

  - is of self cooling type

Types

- Power transformer

  - have ratings above 200 kVA

  - used to step up voltages

  -put in operation during load hours and disconnected during light-load hours

  - designed to have maximum efficiency at or near full load

  - flux density is around 1.5-1.7 Wb/m^2

  - voltage regulation is around 6-10%

  - may be self cooled or forced oil cooled

Types

5. Transformers based on the nature of supply

- Single-phase transformer.
- Three-phase transformer.

6. Transformers for measurement purposes (Instrument transformers)

- Current transformers
  - used to measure high currents of the order of kiloamperes
  - primary has few turns of heavy wire
  - secondary has more turns of fine wire
- Potential transformers
  - used to measure high voltages of the order of kilovolts
  - primary has more turns
  - secondary has few turns

Power & Distribution Transformers -Difference

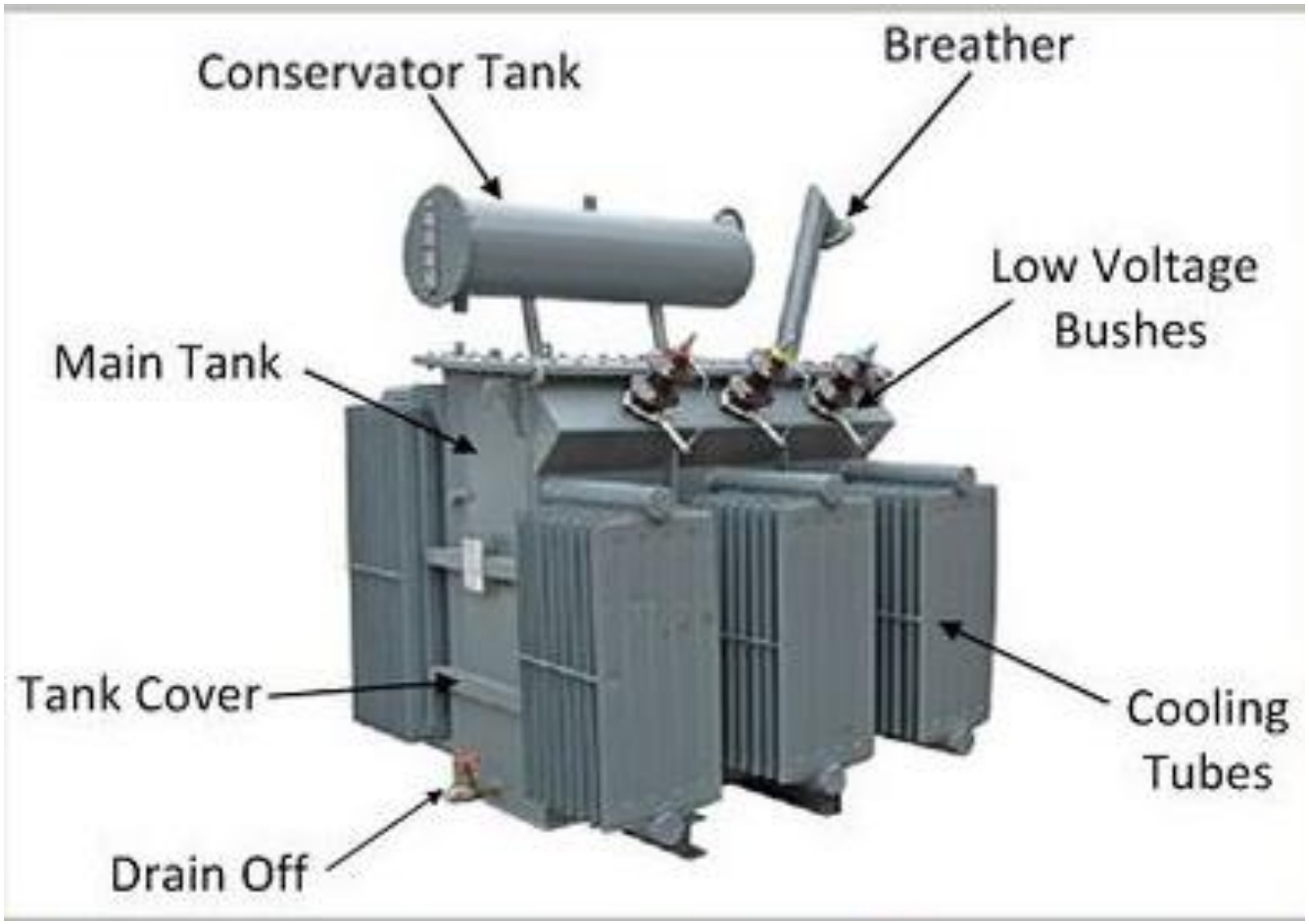
Sl. No.	Power Transformer	Distribution Transformer
1.	used in transmission network of higher voltages	used in the distribution network of lower voltages.
2.	Larger in size as compared of distribution transformers	Smaller in size
3.	Designed for maximum efficiency of 100%	Designed for 50-70% efficiency
4	Used in generating stations and transmission substations	Used in distribution stations, also for industrial and domestic purposes
5	Copper and iron losses take place throughout the day	Iron losses take place for 24 hours and copper losses are based on load cycle
6	In power transformer the load fluctuations are very less	Load fluctuations are very high
7	In power transformer flux density is higher	As compared to power transformer the flux density is lower in distribution transformer

Construction

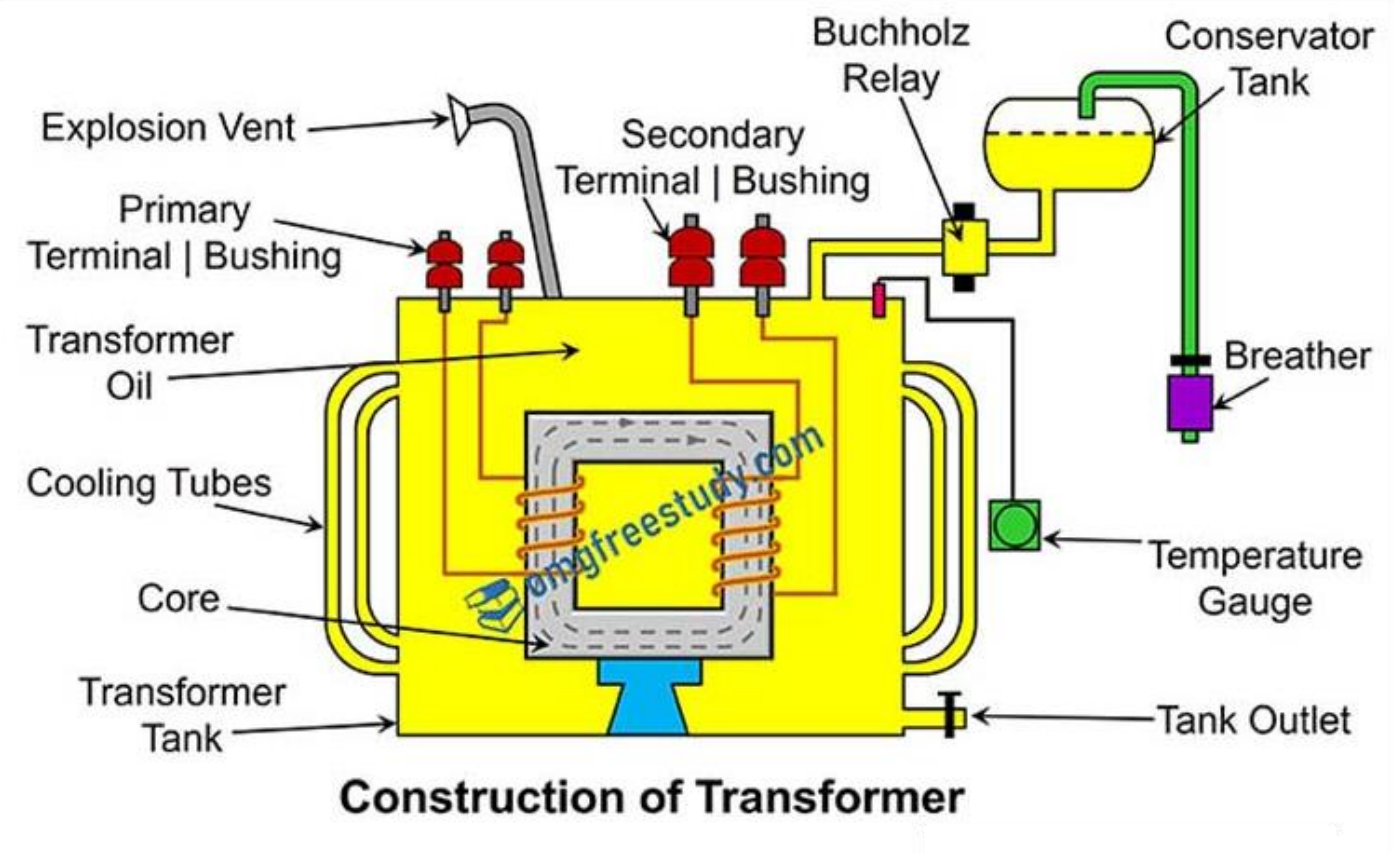
Main parts:-

- 2 coils
- Laminated steel core
- Container for placing assembled core and the windings – transformer tank
- A medium for insulating the core and the windings – transformer oil
- Bushings – for bringing out terminals of the windings from the tank
- Conservator – small tank placed on the top of the main tank- take up contraction and expansion of oil without allowing it to come in contact with outside air
- Breather- collects moisture content from the transformer oil

Construction



Construction



Construction

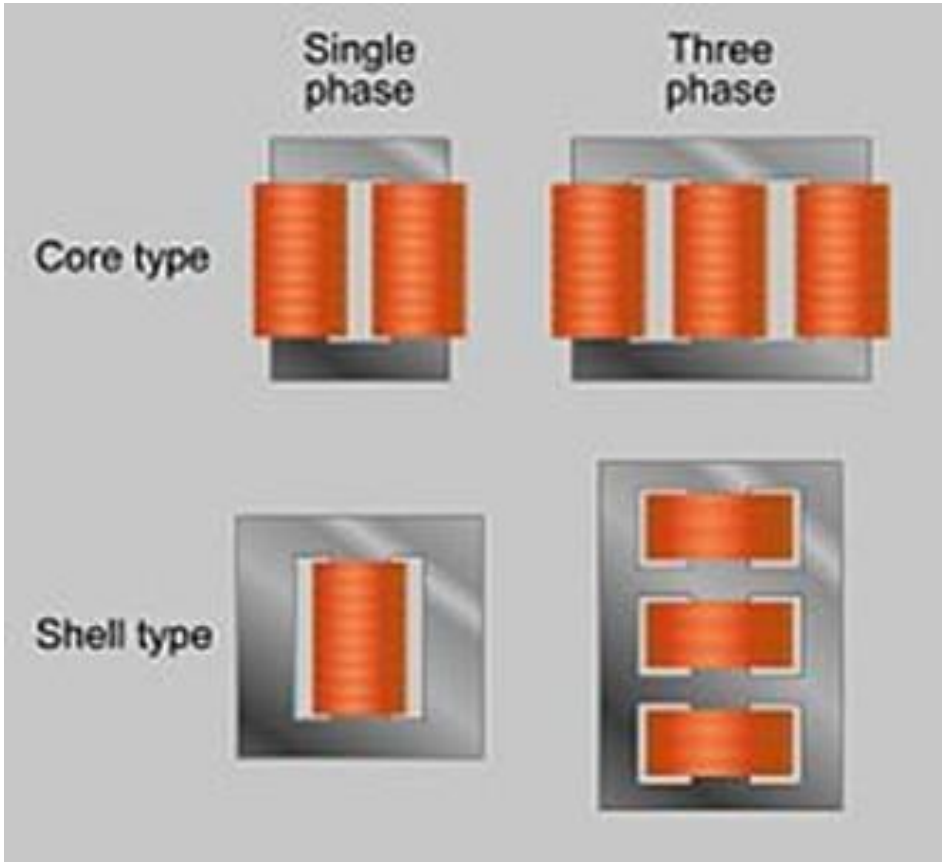
Core:-

- Provides a continuous magnetic path
- Made of sheet steel laminations
- Steel is of high silicon content
- This is for high permeability and low hysterisis loss
- Laminating the core reduces the eddy current loss
- Laminations are insulated from each other by a light coat of varnish
- Thickness of laminations – 0.35mm for a frequency of 50 Hz

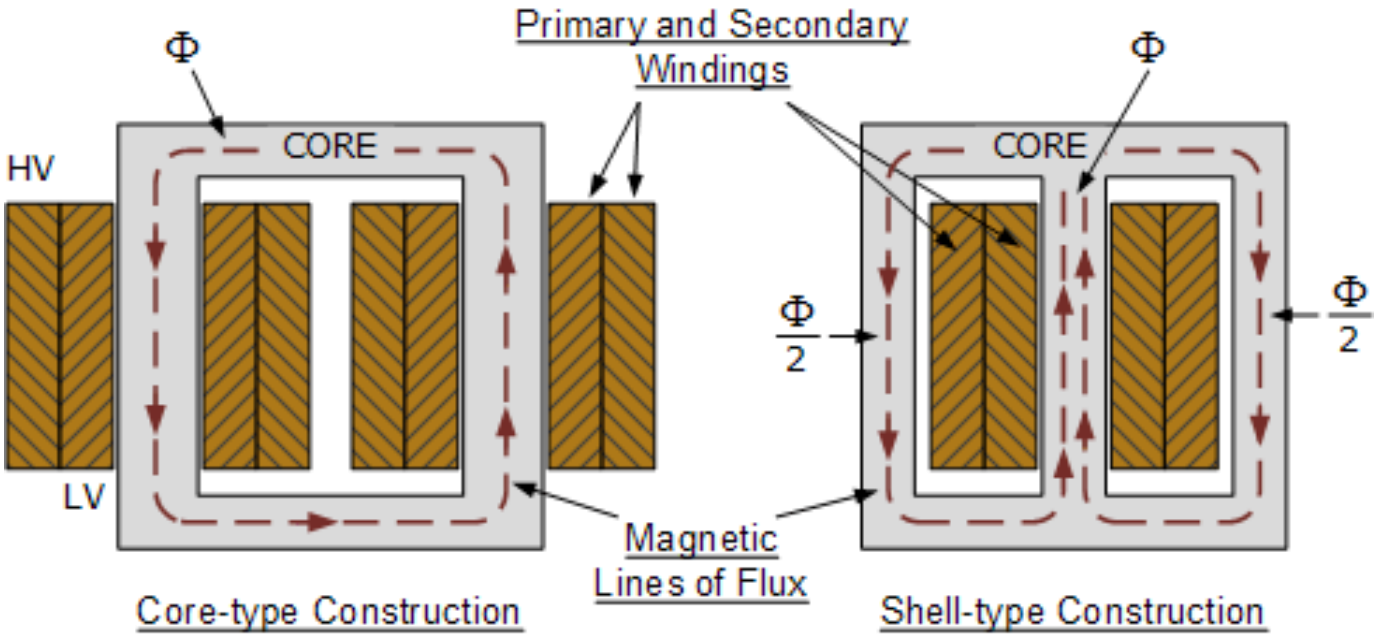
Construction

- 2 types- core type & shell type
- Core type – winding surrounds the core
- Shell type- core surrounds the winding
- In an actual transformer, the 2 windings are interleaved to reduce the leakage flux
- In an actual core type transformer – half the number of both the primary winding & secondary winding are placed on each limb

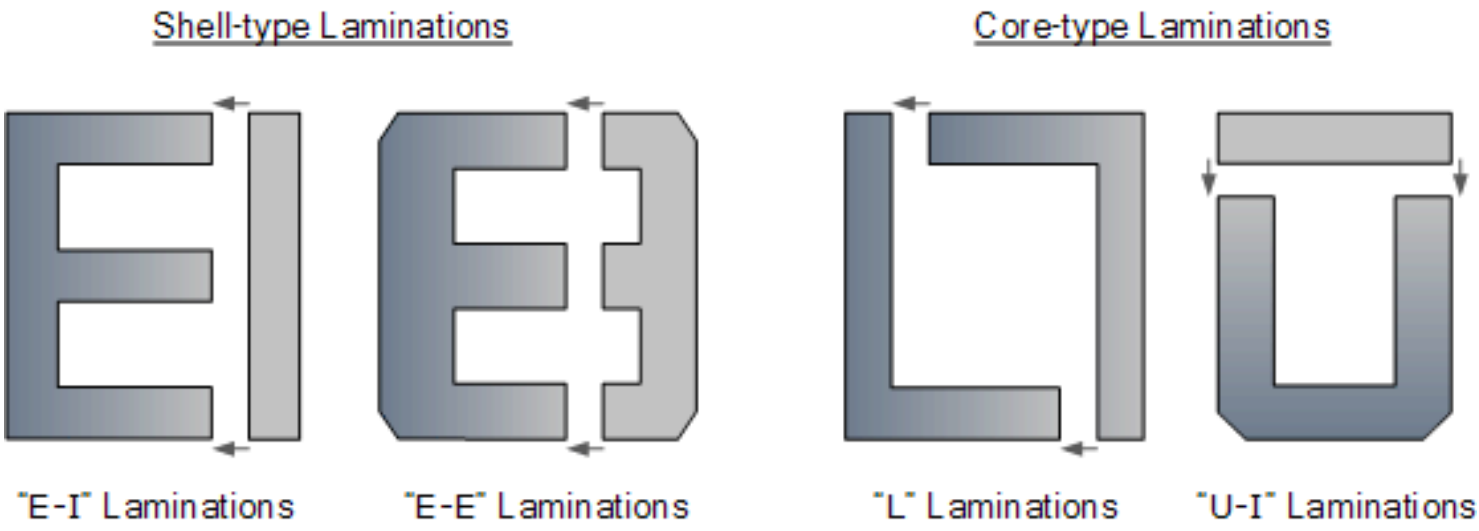
Construction



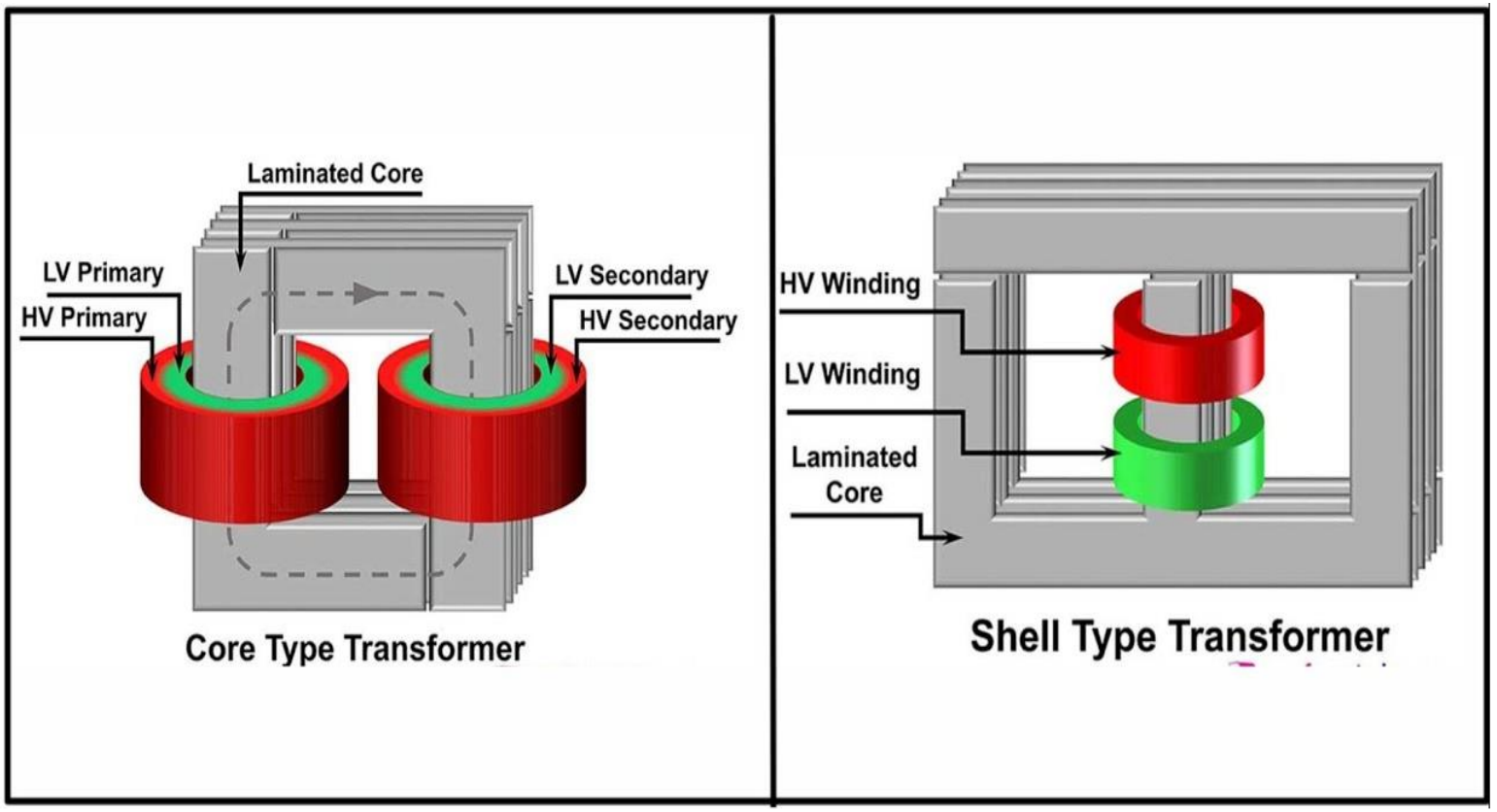
Construction



Construction



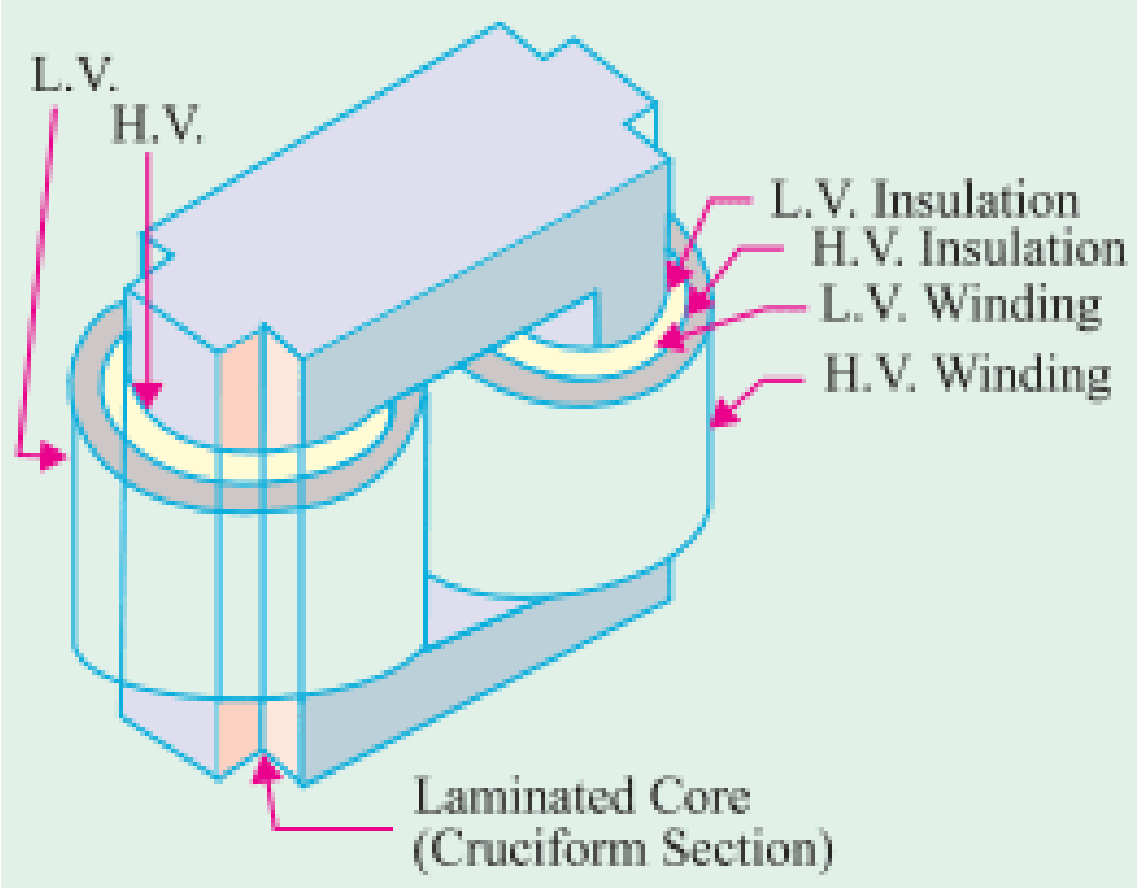
Core Type Transformers



Core Type Transformers

- Coils are of form-wound and cylindrical type
- Core are of U-I laminations
- Small size core type transformers-rectangular core with cylindrical coils either circular or rectangular
- Large size core type transformers-round or cylindrical coils over a cruciform core section
- Circular cylindrical coils – high mechanical strength
- LV windings are placed near the core – because it is easy to insulate

Core Type Transformers



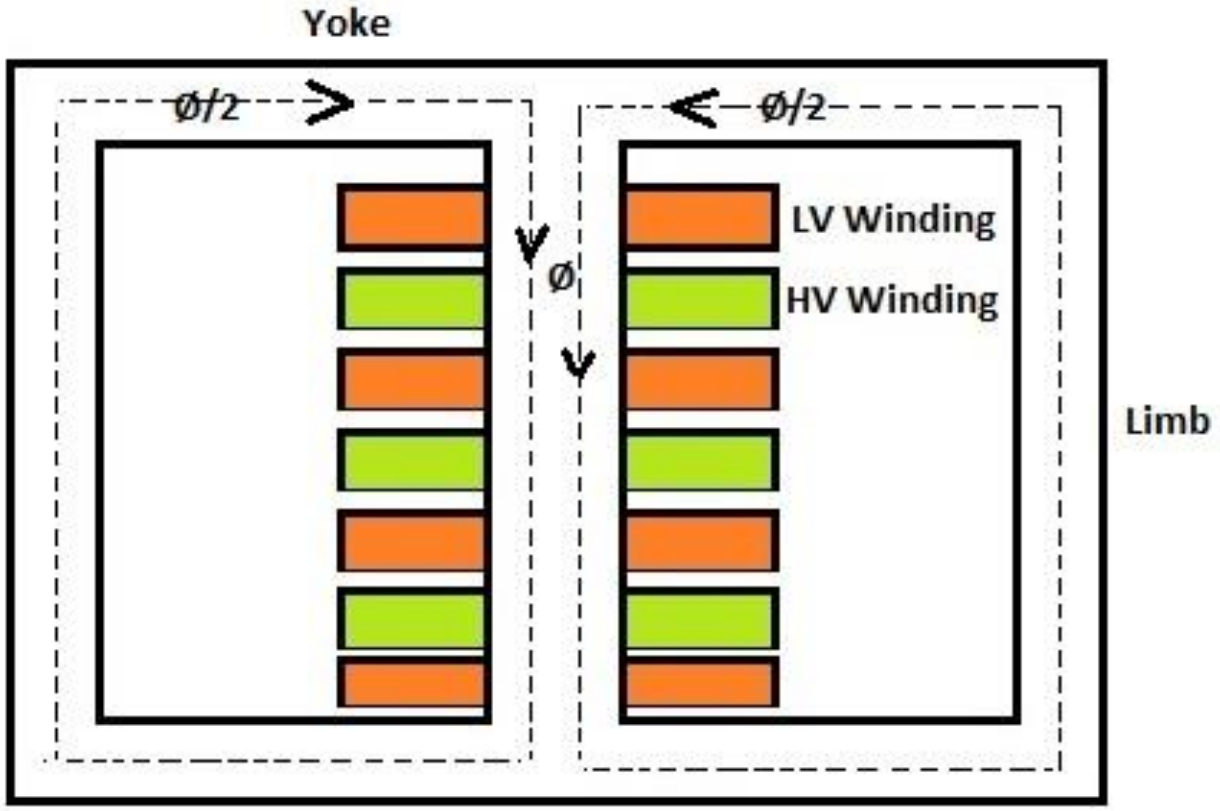
Core Type Transformers

- 2 windings are arrange as concentric coils- called concentric winding
- Easier to dismantle for repair
- Natural cooling is better as the coils are exposed

Shell Type Transformers

- Both the windings are wound on the central limb
- Windings are put in the form of a sandwich-called sandwich or disc winding
- Mainly E-I laminations are used
- Gives better support against electromagnetic forces between the conductor
- Natural cooling is poor due to embedding of the coils
- Preferred for very high voltage transformers

Shell Type Transformers



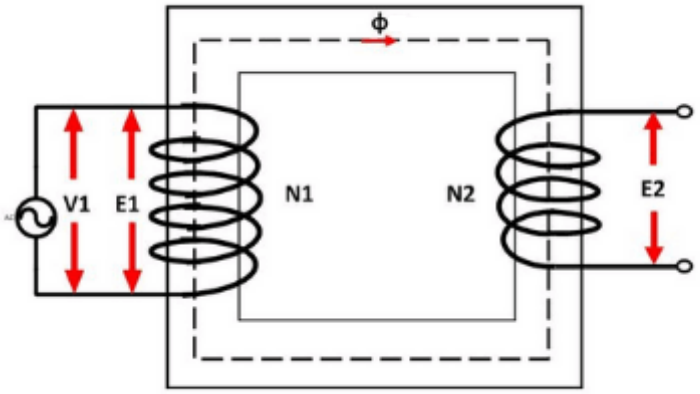
Core & Shell Type - Differences

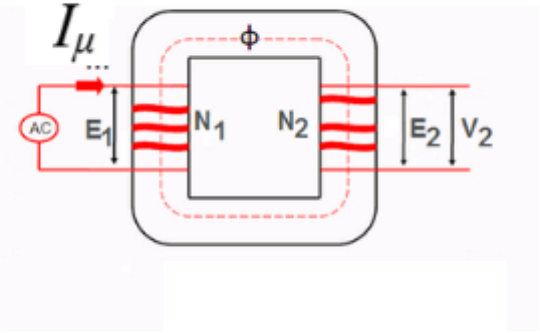
Sl. No.	Core Type	Shell Type
1	the windings surround the core of transformer	the magnetic core surrounds the windings of the transformer.
2	U and I shaped laminations are used	made up of either U and T shaped laminations or E and I shaped laminations.
3	consists of a single magnetic circuit	consists of two magnetic circuits
4	The core loss is more because the total magnetic flux flows through the entire core.	The core loss is comparatively less because half of the total flux flows in the entire core.
5	used for high voltage and high power transformers.	Used for low voltage and low power applications.

Ideal Transformer

- No losses
- Windings have no ohmic resistance
- Whole of the magnetic flux is confined to the core; i.e., no magnetic leakage
- No I^2R loss and no core loss
- It consists of 2 purely inductive coils wound on a loss-free core
- Permeability of the core is infinite

Ideal Transformer





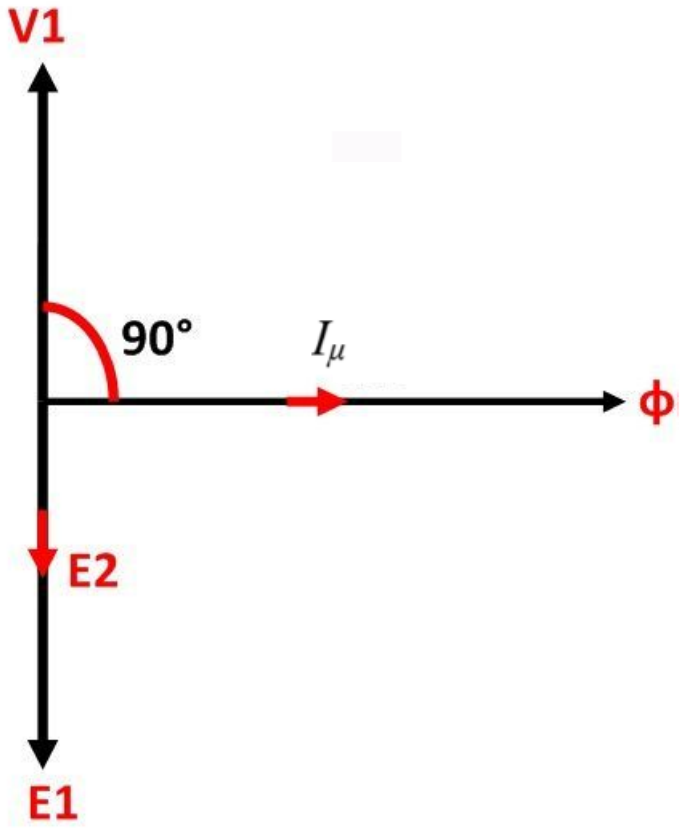
Ideal Transformer

- Primary is connected to an alternating voltage V_1 and secondary is open
- Causes a current to flow in the primary
- Primary draws only the magnetising current(I_μ) because the primary coil is inductive and secondary is open
- I_μ-magnetises the core, small in magnitude, lags V_1 by 90°, produces the alternating flux Φ and is proportional to it.
- I_μ is in phase with Φ because the permeability of the magnetic circuit is assumed to be constant

Ideal Transformer

- Produces a self induced emf E_1
- At every instant E_1 is equal and in opposition to V_1.
- E_1 is also called **counter emf** or **back emf of the primary**
- emf is induced in the secondary – called mutually induced emf (E_2).
- E_2 **-** in antiphase with V_1 , proportional to rate of change of flux & number of turns in the secondary

Ideal Transformer



EMF Equation

$Let\ N_1 = number\ of\ turns\ in\ the\ primary$

$N_2 = number\ of\ turns\ in\ the\ secondary$

$f = frequency\ of\ AC\ input; Hz$

$\Phi_m = maxium\ value\ of\ flux\ in\ the\ core; Wb$

$= maximum\ flux\ density in\ the\ core\ in\ Wb/m^2$

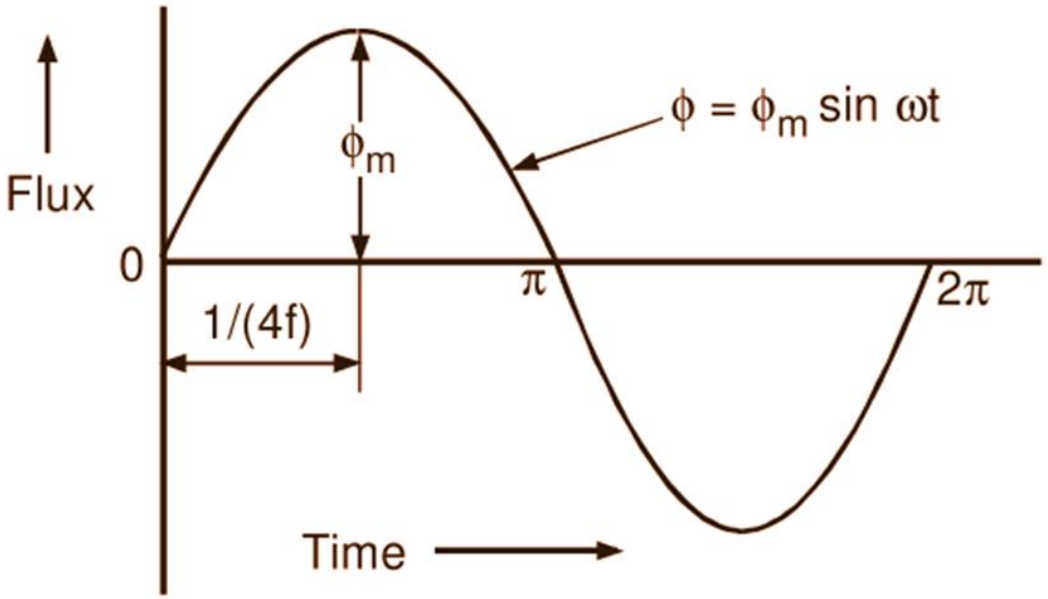
$x\ area\ of\ cross\text{-}section\ of\ the\ core\ in\ m^2 = B_m\ x\ A$

$$Average\ rate\ of\ change\ of\ flux = \frac{\Phi_m}{1/4f} = 4f\Phi_m; \text{volts}$$

$Induced\ emf\ in\ volts = rate\ of\ change\ of\ flux/turn$

$$average\ emf/turn = 4f\Phi_m; \text{volts}$$

EMF Equation



EMF Equation

flux Φ varies sinusoidally, so form factor is 1.11

rms value of induced emf/turn = average emf/turn x form factor

$$= 4f\Phi_m \, x \, 1.11 = 4.44f\Phi_m$$

rms value of induced emf in the primary winding =

$$E_1 = 4.44f\Phi_m N_1 = 4.44fN_1 B_m A$$

rms value of induced emf in the secondary winding =

$$E_2 = 4.44f\Phi_m N_2 = 4.44fN_2 B_m A$$

$$\text{i.e., emf/turn} = \frac{E_1}{N_1} = \frac{E_2}{N_2} = 4.44f\Phi_m$$

emf/turn is same for both primary & secondary

For an ideal transformer on no-load

$V_1 = E_1$ & $E_2 = V_2$, where V_2 is the terminal voltage

$$\frac{E_2}{E_1} = \frac{N_2}{N_1} = K;\ \text{voltage transformation ratio}$$

$$\frac{N_1}{N_2} = \text{turns ratio}$$

Step-up transformer : -if $N_2 > N_1$ i.e., $K > 1$

Step-down transformer : -if $N_2 < N_1$ i.e., $K < 1$

for an ideal transformer,

$$V_1 I_1 = V_2 I_2 \qquad \text{i.e.,} \frac{I_2}{I_1} = \frac{V_1}{V_2} = \frac{1}{K}$$

where I_1 = primary current & I_2 = secondary current

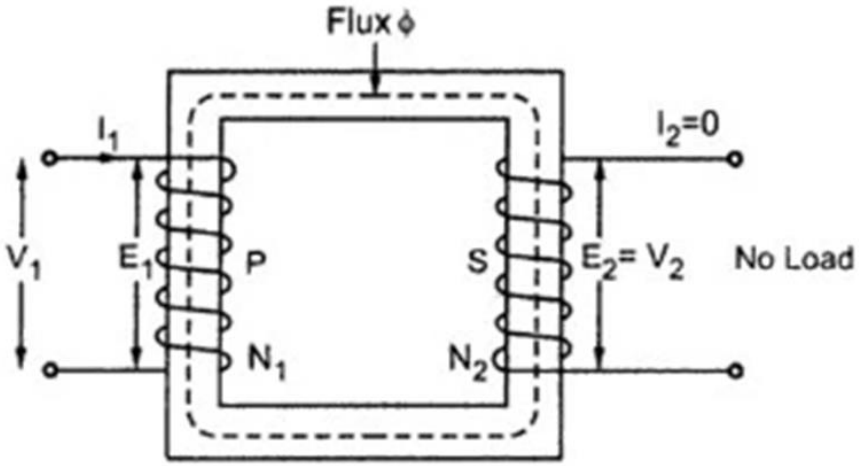
Practical Transformer on No-load

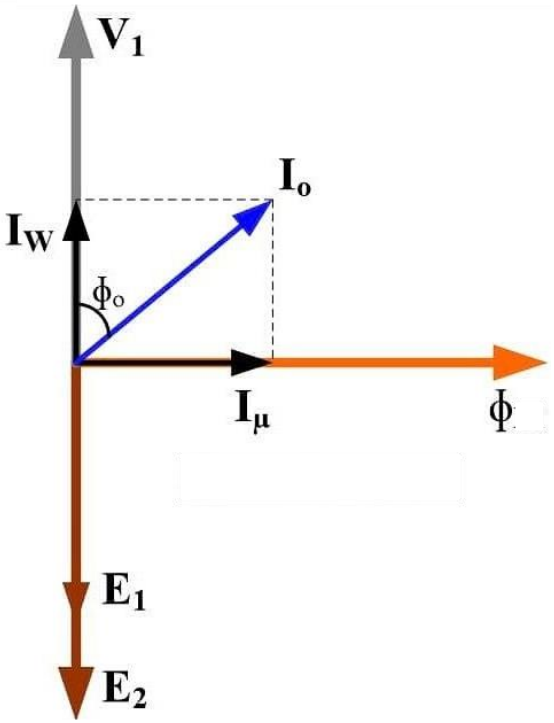
- No-load – secondary circuit is open, i.e., $I_2 = 0$
- actual transformer - iron loss in the core and copper loss in both primary and secondary windings
- Primary current has to supply

  – supply iron loss in the core

  - small copper loss in the primary [$I_2 = 0$; so no copper loss in the secondary winding]
- No-load primary current I_0 lags V_1 by an angle Φ_0, $\Phi_0 < 90^0$

Practical Transformer on No-load





Practical Transformer on No-load

$No\text{-}load\ input\ power\ , W_0 = V_1 I_0 \cos\phi_0$

$where\ \cos\phi_0 = no-load\ primary\ power\ factor$

$2\ components\ of\ I_0\ :\text{-}$

I_w – *active or working or iron loss component*

- *in phase with* V_1
- *supplies iron loss and primary copper loss*

$= I_0 \cos\phi_0$

I_μ – *magnetising component*

- *in quadrature with* V_1
- *produces alternating flux in the core*

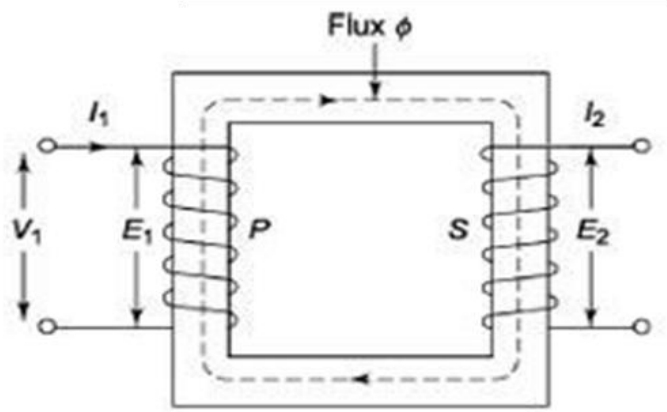
$= I_0 \sin\phi_0$

$I_0 = \sqrt{I_\mu^{\ 2} + I_w^{\ 2}}$

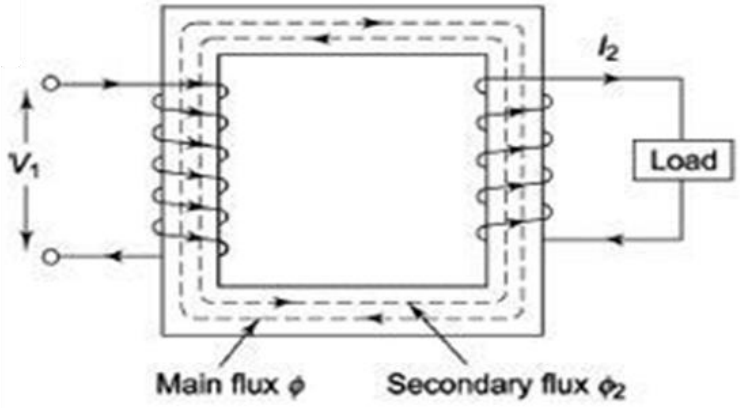
Practical Transformer on No-load

- I_0 is very small , about 1% of the full load current
- Since I_0 is very small , the no-load primary copper loss is very small
- i.e., **no-load primary input power is equal to the iron loss in the transformer**

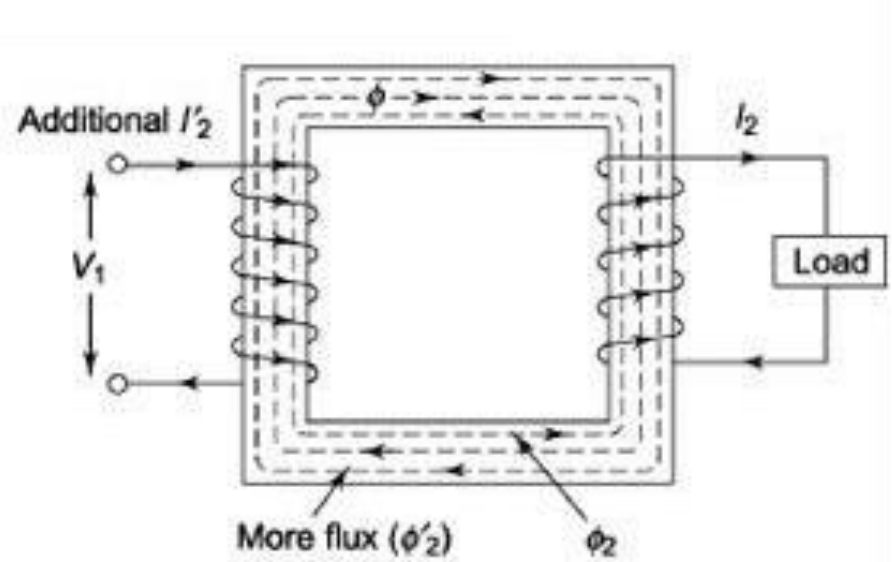
Practical Transformer on Load



(a) No-load condition



(b) ϕ_2 Opposes ϕ



(c) Primary draws more current

Actual Transformer on Load (Flux constant)

Practical Transformer on Load

- When loaded, I_2 is set up
- Magnitude & phase of I_2 wrt V_2 – determined by the characteristics of the load
- I_2 in phase with V_2 – when load is resistive
- I_2 lags V_2 – when load is inductive
- I_2 leads V_2 – when load is capacitive
- I_2 sets up an mmf $N_2 I_2$ –and also its own flux Φ_2
- Φ_2 is in opposition to main flux Φ
- Φ *is due to* I_0

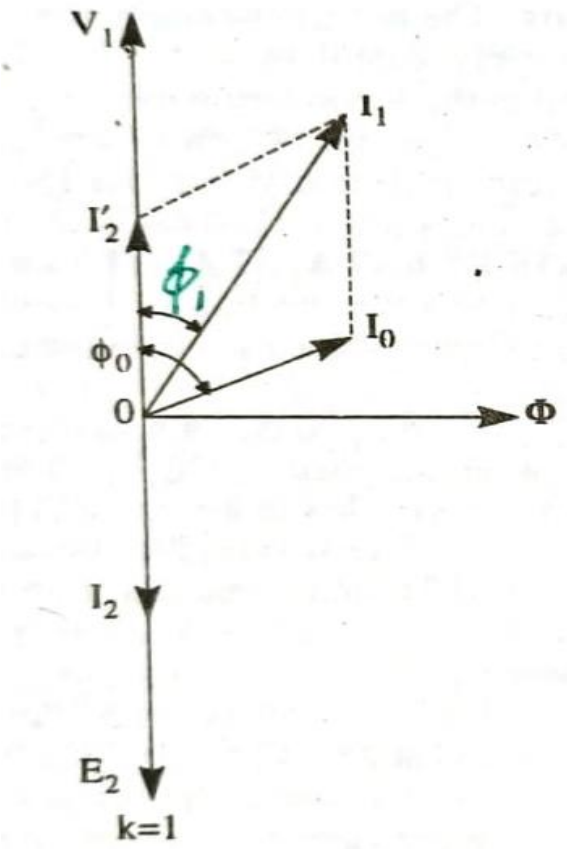
Practical Transformer on Load

- $N_2 I_2$ –called demagnetising ampere turns
- Φ_2 weakens Φ momentarily
- Hence E_1 tends to be reduced.
- V_1 gains upper hand and causes more current to flow in the primary
- Let I_2' be the additional primary current – called load component of primary current
- I_2' is in antiphase with I_2
- I_2' sets up an additional primary mmf $N_1 I_2'$ and its own flux Φ_2'

Practical Transformer on Load

- Φ_2' is in opposition to Φ_2 and in the same direction of main flux Φ
- Φ_2' is equal to Φ_2 and they cancel each other
- Magnetic effects of I_2 is neutralized by additional primary current I_2'
- **Whatever be the load conditions the net flux in the core is same as that at no-load**
- **Therefore core loss is also practically same under all load conditions**

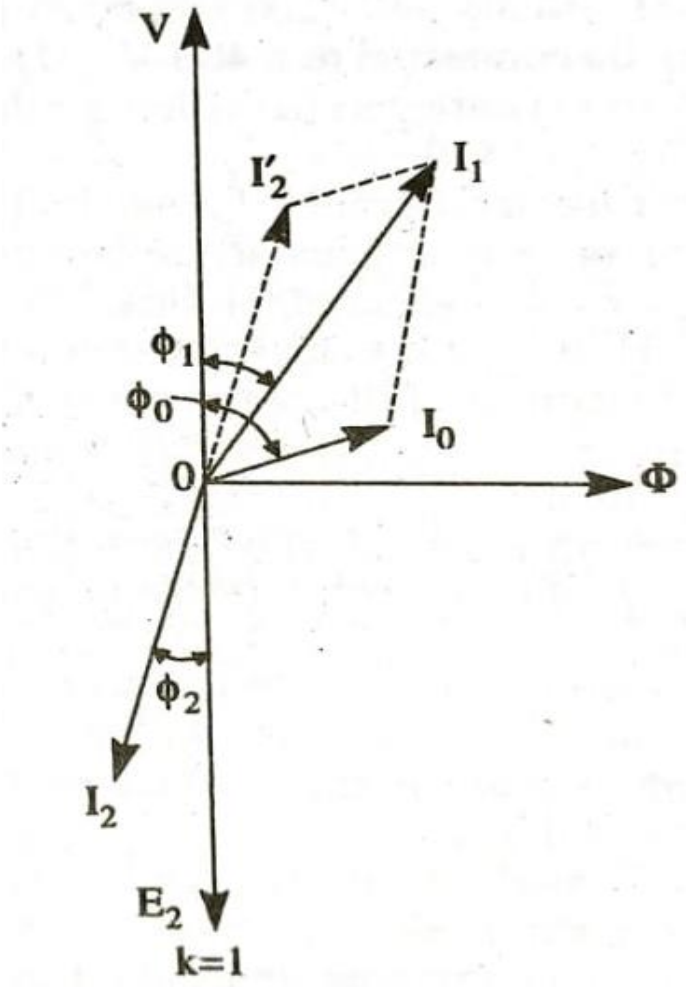
Phasor Diagram of Transformer (with no winding resistance) on Resistive Load-



Transformer (with no winding resistance) on Resistive Load- Phasor Diagram

- Voltage transformation ratio is assumed to be unity
- For resistive load I_2 will be in phase with E_2
- I_2' is antiphase with I_2 and equal in magnitude as K=1
- I_1 –vector sum of I_0 and I_2'
- I_1 lags V_1 by an angle Φ_1

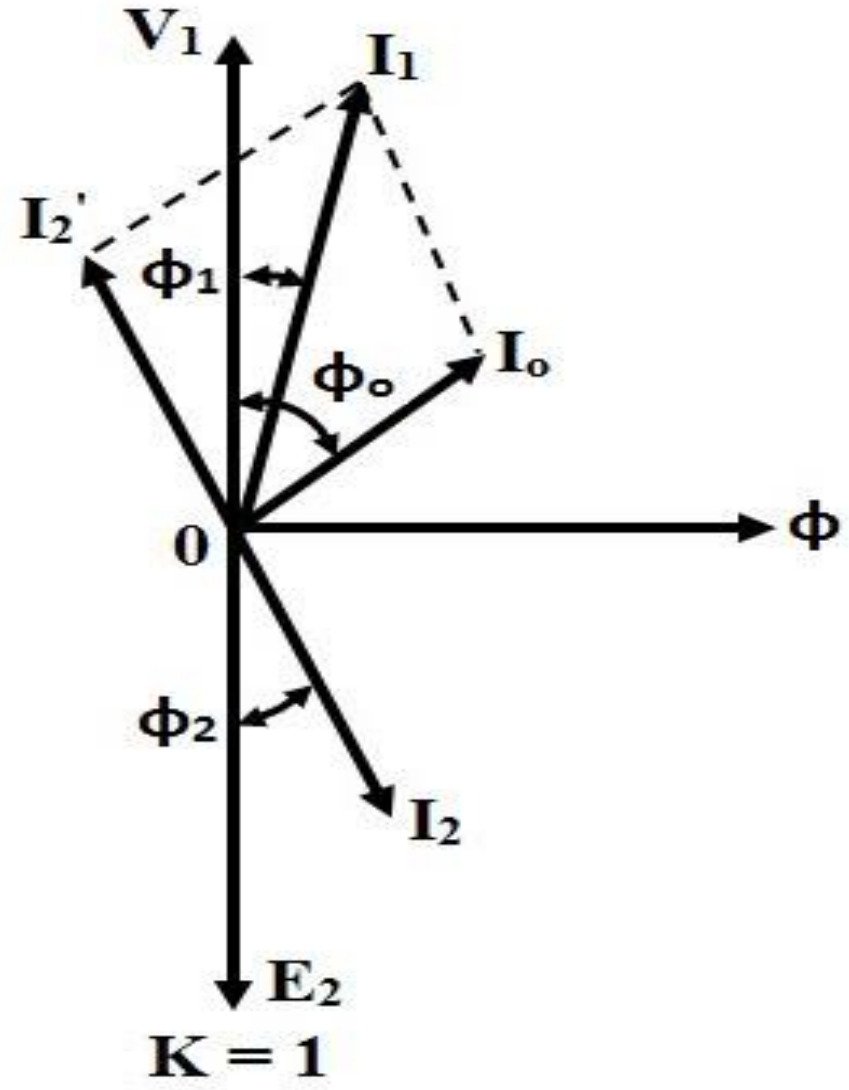
Transformer (with no winding resistance) on Inductive Load- Phasor Diagram



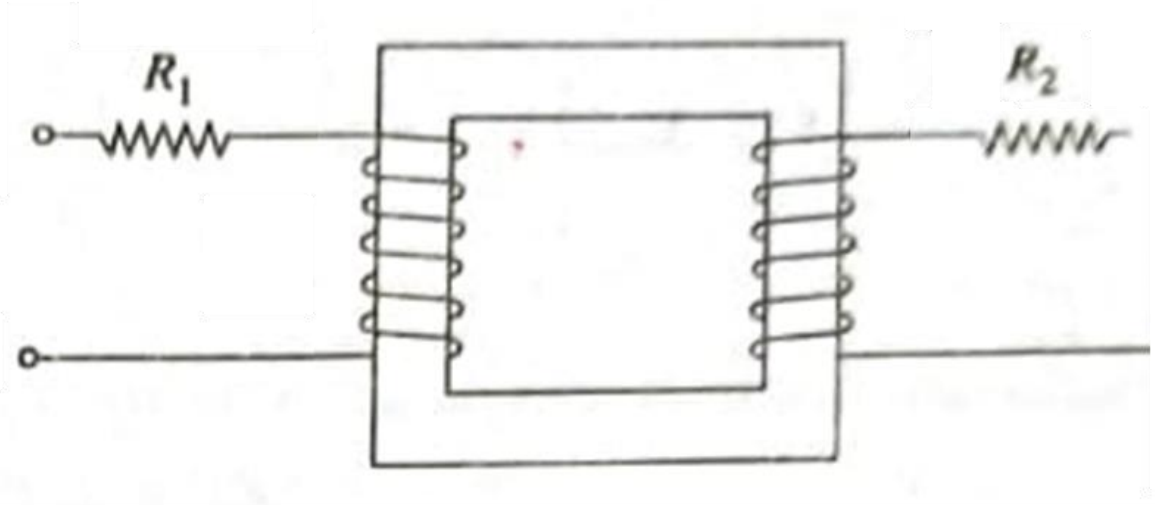
Transformer (with no winding resistance) on Inductive Load

- I_2 lags V_2 by an angle Φ_2
- I_2' is antiphase with I_2 and equal in magnitude as K=1
- I_1 –vector sum of I_0 and I_2'
- I_1 lags V_1 by an angle Φ_1
- Φ_1 is slightly greater than Φ_2.

Transformer (with no winding resistance) on Capacitive Load- Phasor Diagram



Transformer with Resistance & No Magnetic Leakage



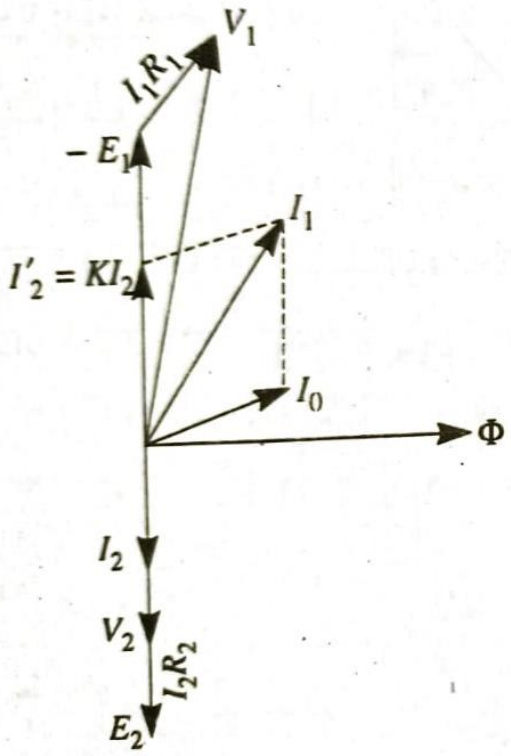
Transformer with Resistance & No Magnetic Leakage

Due to the winding resistance there will be some voltage drop in the 2 windings

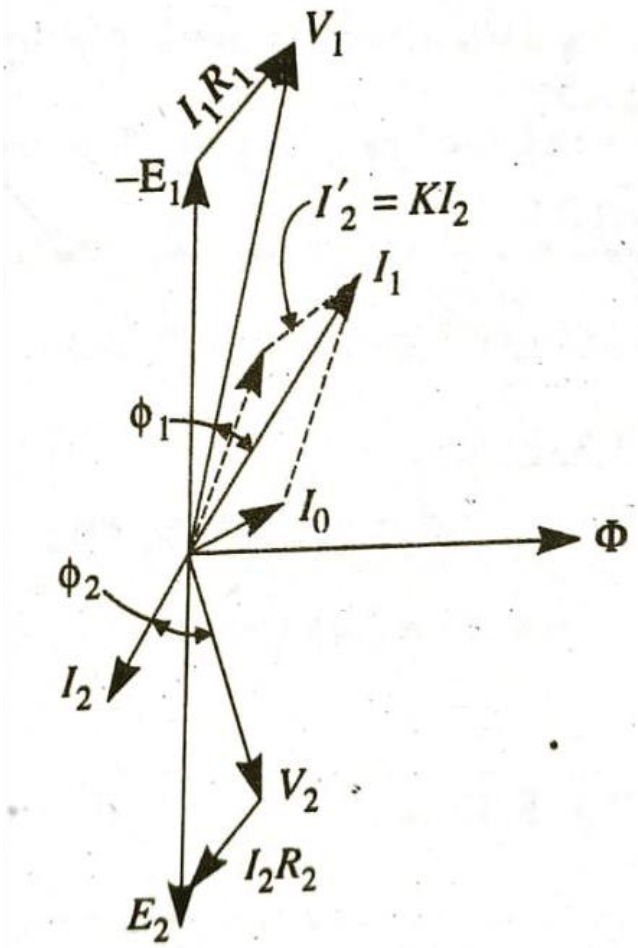
$$E_1 = V_1 - I_1 R_1$$

$$V_2 = E_2 - I_2 R_2$$

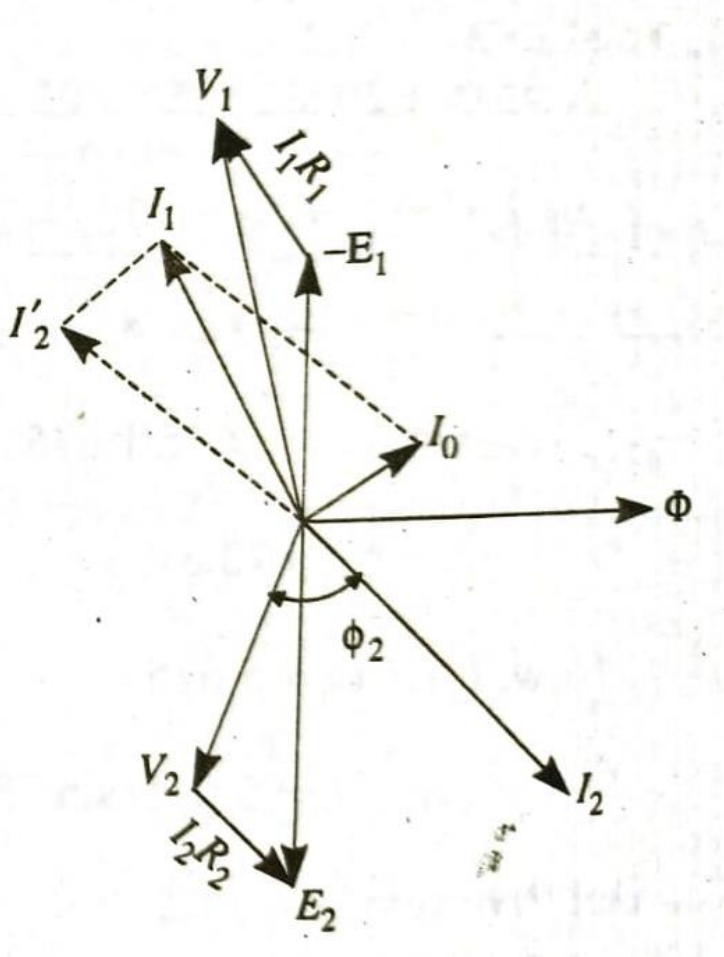
Transformer with Resistance & No Magnetic Leakage- Resistive Load



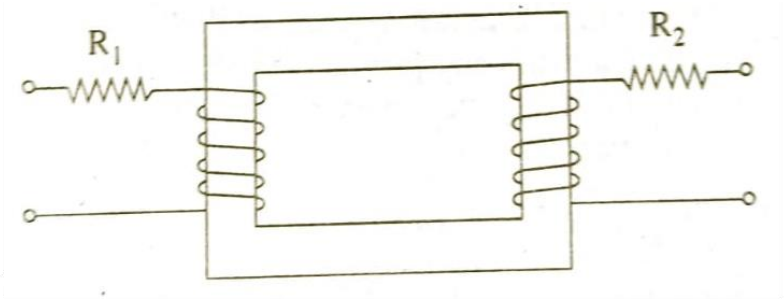
Transformer with Resistance & No Magnetic Leakage- Inductive Load



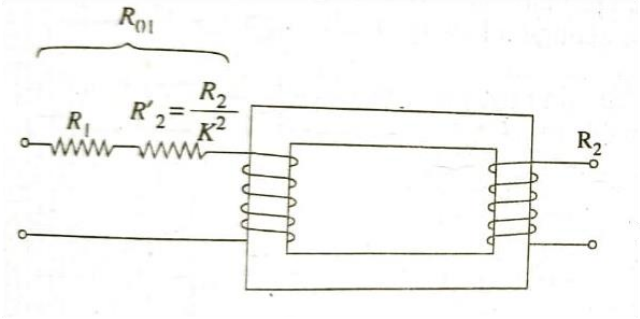
Transformer with Resistance & No Magnetic Leakage-Capacitive Load



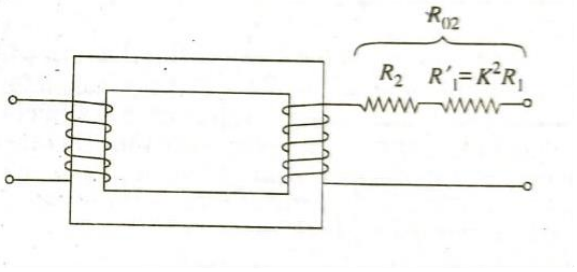
Transformer- Equivalent Resistance



Transformer with primary & secondary resistances shown externally



Transformer with all resistances concentrated in the primary



Transformer with all resistances concentrated in the secondary

Equivalent Resistance

- Resistance of any of the windings can be transferred to any one of the 2 windings
- This makes the calculation simple
- Any resistance transferred to the primary is divided by K^2.
- i.e., R_2 in secondary is equivalent to $R_2' = R_2 / K^2$ in primary, where R_2' is the equivalent secondary resistance referred to primary
- Any resistance transferred to the secondary is multiplied by K^2.
- i.e., R_1 in primary is equivalent to $R_1' = R_1 K^2$ in secondary, where R_1' is the equivalent primary resistance referred to secondary

Equivalent Resistance

Copper loss in secondary winding $= \mathrm{I}_2^2 R_2$

$R_2^{'}$ *is the equivalent resistance in the primary*

which would've caused the same loss as R_2 *in secondary*

$$\mathrm{I}_1^2 R_2^{'} = \mathrm{I}_2^2 R_2 \qquad \text{or } R_2^{'} = \left(\frac{\mathrm{I}_2}{\mathrm{I}_1}\right)^2 R_2 = R_2 / K^2$$

$$\because \frac{I_2}{I_1} = \frac{1}{K}$$

Equivalent Resistance

equivalent primary resistance referred to secondary ,

$$R_1^{'} = K^2 R_1$$

equivalent secondary resistance referred to primary ,

$$R_2^{'} = R_2 / K^2$$

equivalent resistanace of the transformer referred to primary,

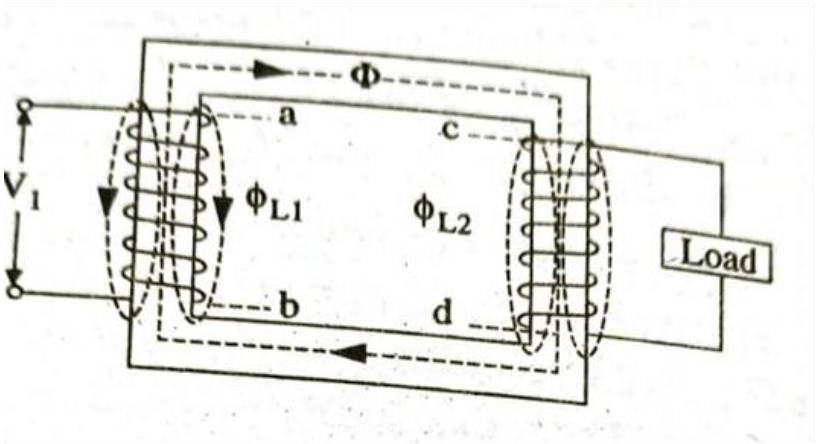
$$R_{01} = R_1 + R_2^{'} = R_1 + R_2 / K^2$$

equivalent resistanace of the transformer referred to secondary,

$$R_{02} = R_2 + R_1^{'} = R_2 + K^2 R_1$$

Magnetic Leakage

- In the case of an ideal transformer all the flux linking with the primary also links the secondary winding
- In actual transformer it is not so
- Φ_{L1}(or Φ_{L2}) completes the magnetic circuit passing through air rather than around the core



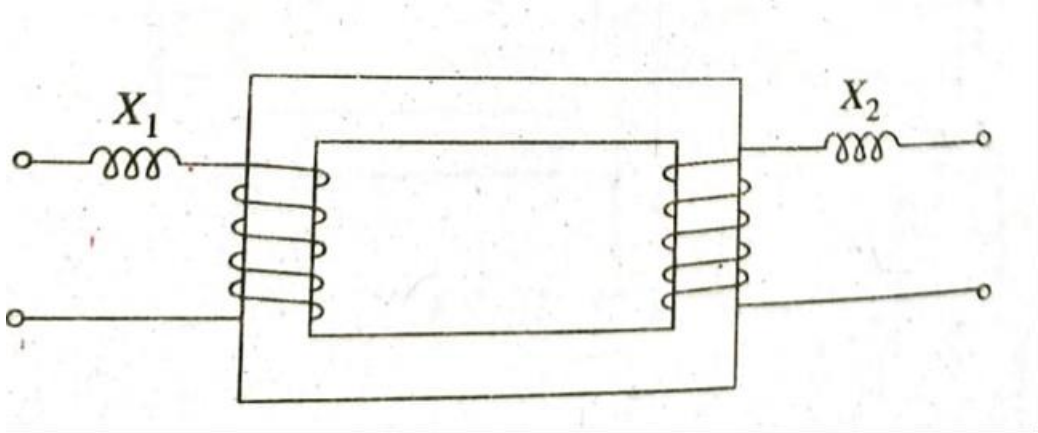
Magnetic Leakage

- Leakage flux is produced when the mmf due to primary ampere turns lying between points ‘a’ and ‘b’ acts along the leakage paths
- Φ_{L1} links with primary winding alone
- Φ_{L1} is in phase with I_1 .
- Φ_{L1} induces an emf e_{L1} in the primary
- secondary ampere turns between points ‘c’ and ‘d’ sets up Φ_{L2} .
- Φ_{L2} links with secondary winding alone
- Φ_{L2} is in phase with I_2 .
- Φ_{L2} induces an emf e_{L2} in the secondary

Magnetic Leakage

- At no load and light loads, leakage flux is small as the ampere turns are small
- When load is increased the currents will set up large mmfs which increase the leakage flux
- The effect of leakage flux can be represented by an inductance in series with the winding
- A transformer with magnetic leakage – ideal transformer with inductive coils in series with the winding

Magnetic Leakage



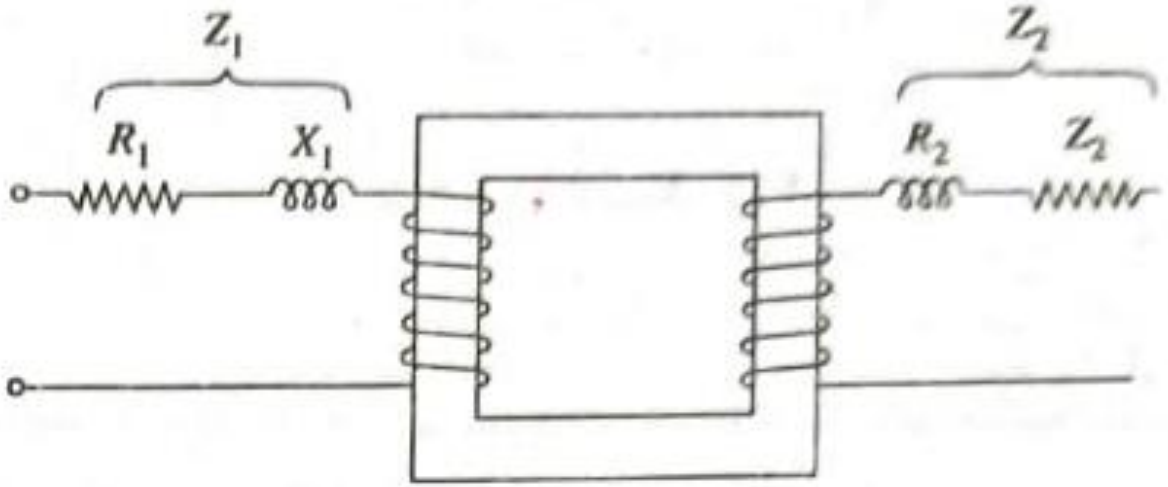
Magnetic Leakage

- Primary leakage reactance, $X_1 = e_{L1}/I_1$
- Secondary leakage reactance, $X_2 = e_{L2}/I_2$

Leakage flux:-

- Links with one winding only
- Does not contribute to the transfer of energy from primary to secondary
- Interleaving of the primary & the secondary windings minimises the leakage flux

Transformer with Resistance & Magnetic Leakage



Transformer with Resistance & Magnetic Leakage

$$\textit{Primary impedance}, Z_1 = \sqrt{(R_1^2 + \mathrm{X}_1^2)}$$

$$\textit{Secondary impedance}, Z_2 = \sqrt{(R_2^2 + \mathrm{X}_2^2)}$$

$$V_1 = E_1 + I_1(R_1 + jX_1) = E_1 + I_1 Z_1$$

$I_1 X_1$ *is usually 1 or 2% of* V_1

$$E_2 = V_2 + I_2(R_2 + jX_2) = E_2 + I_2 Z_2$$

Leakage Reactance

- Leakage reactance of any of the windings can be transferred to any one of the 2 windings
- Any reactance transferred to the primary is divided by K^2.
- i.e., R_2 in secondary is equivalent to $R_2' = R_2 / K^2$ in primary, where R_2' is the equivalent secondary resistance referred to primary
- Any resistance transferred to the secondary is multiplied by K^2.
- i.e., R_1 in primary is equivalent to $R_1' = R_1 K^2$ in secondary, where R_1' is the equivalent primary resistance referred to secondary

Equivalent Leakage Reactance

Leakage reactances can also be transferred from one winding to another.

equivalent primary reactance referred to secondary ,

$$X_1' = K^2 X_1$$

equivalent secondary reactance referred to primary ,

$$X_2' = X_2 / K^2$$

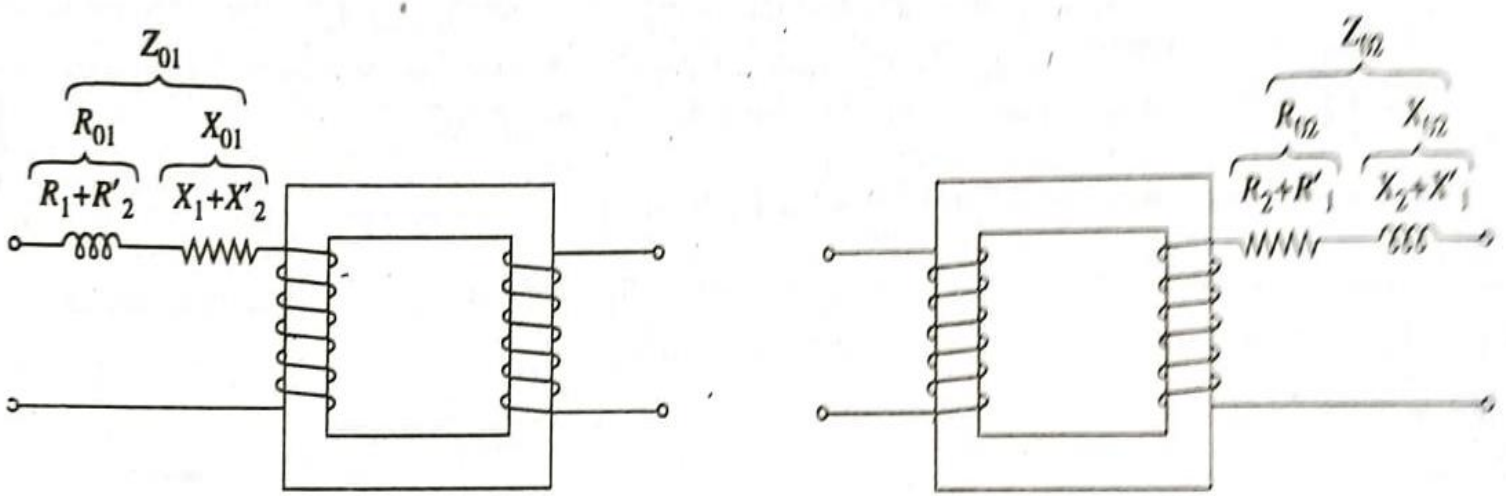
equivalent reactance of the transformer referred to primary,

$$X_{01} = X_1 + X_2' = X_1 + X_2 / K^2$$

equivalent reactance of the transformer referred to secondary,

$$X_{02} = X_2 + X_1' = X_2 + K^2 X_1$$

Transformer with Resistance & Magnetic Leakage- Total Impedance



Transformer with Resistance & Magnetic Leakage- Total Impedance

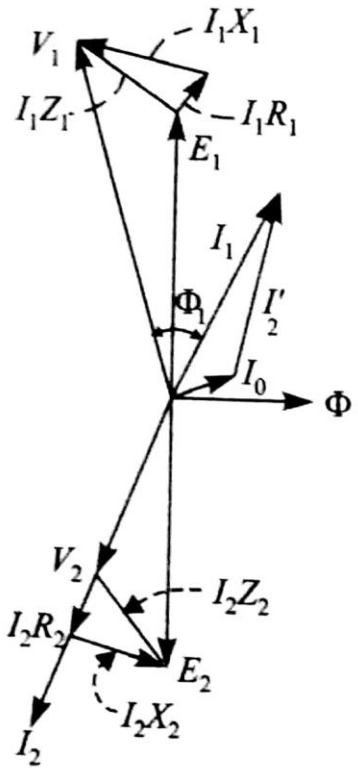
Total impedance of the transformer referred to primary,

$$Z_{01} = \sqrt{(R_{01}^2 + X_{01}^2)}$$

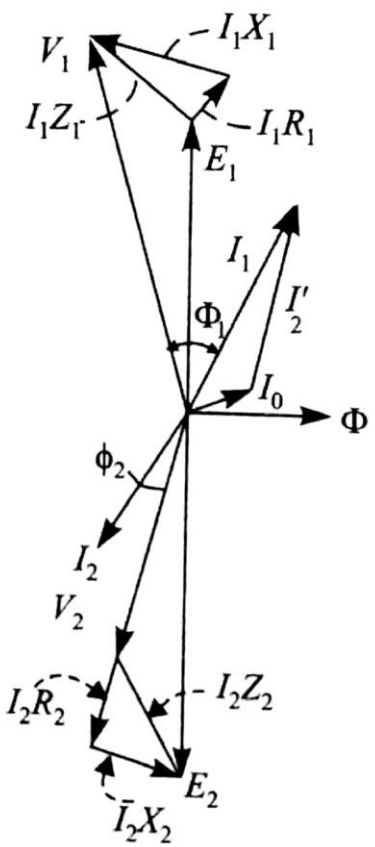
Total impedance of the transformer referred to secondary,

$$Z_{02} = \sqrt{(R_{02}^2 + X_{02}^2)}$$

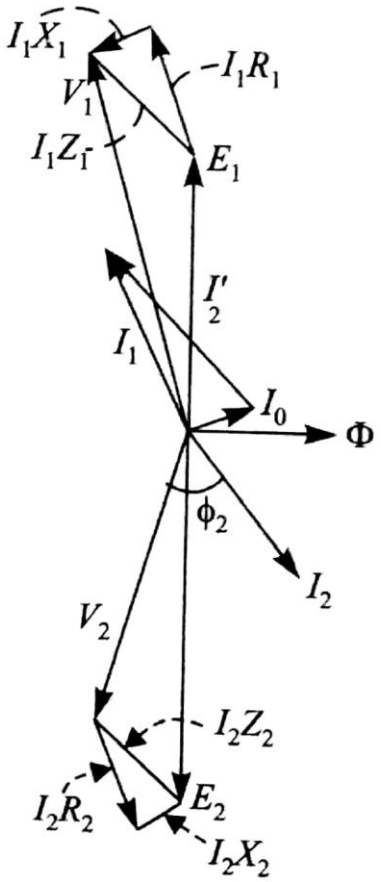
Phasor diagram Transformer with Resistance & Magnetic Leakage



Resistive Load

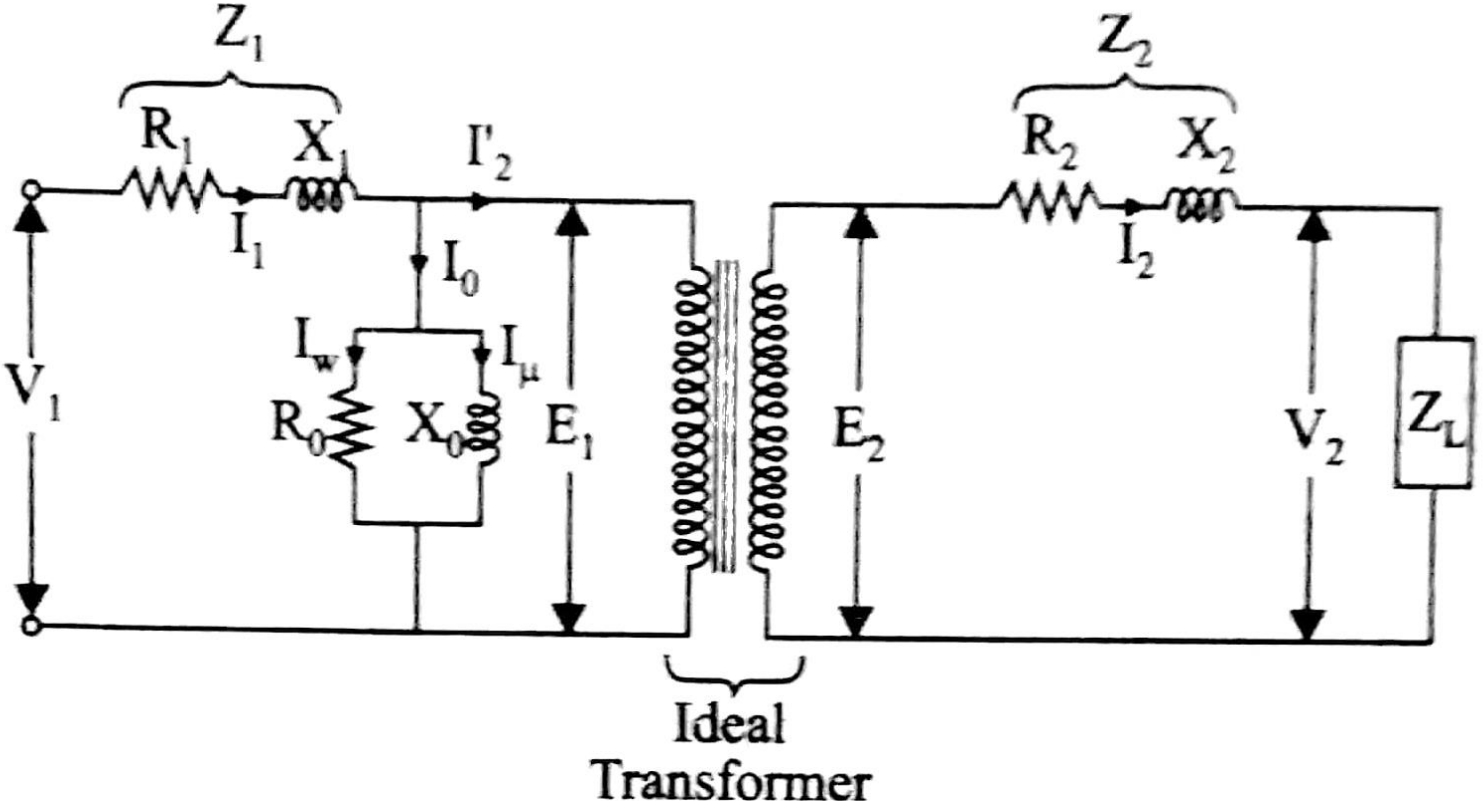


Inductive Load



Capacitive Load

Transformer Equivalent Circuit



Voltage Regulation

- The voltage regulation can be defined in two ways
    - Regulation Down
    - Regulation up.
- Regulation down: This is defined as ” the change in terminal voltage when a load current at any power factor is applied, expressed as a fraction of the no-load terminal voltage”.

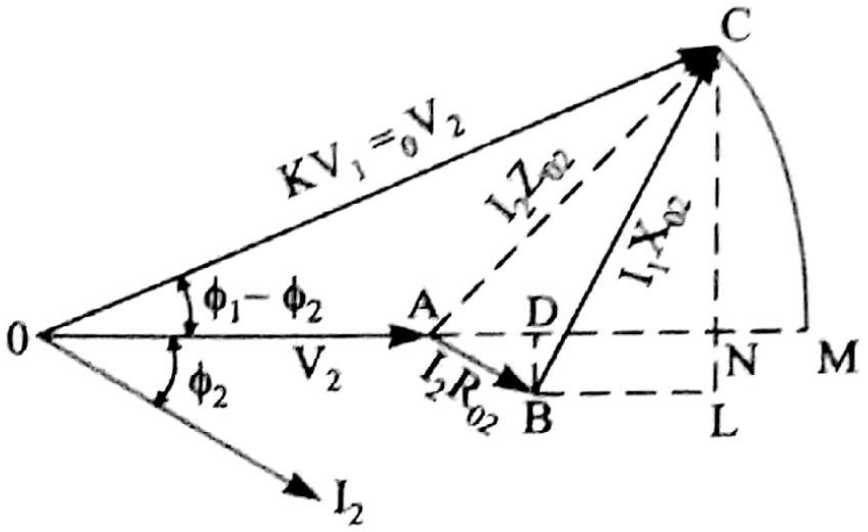
$$Regulation = \frac{|V_{nl}| - |V_l|}{|V_{nl}|}$$

Voltage Regulation

Approximate voltage drop

$$= AN = AD + DN$$

$$= I_2 R_{02} \cos\phi + I_2 X_{02} \sin\phi$$



The approximate voltage drop for *leading* power factor becomes

$$(I_2 R_{02} \cos\phi - I_2 \pm X_{02} \sin\phi)$$

In general, approximate voltage drop is $(I_2 R_{02} \cos\phi \pm I_2 X_{02} \sin\phi)$

It may be noted that approximate voltage drop as referred to primary is

$$(I_1 R_{01} \cos\phi \pm I_1 X_{01} \sin\phi)$$

$$\% \text{ voltage drop in secondary is} = \frac{I_2 R_{02} \cos\phi \pm I_2 X_{02} \sin\phi}{{}_0V_2} \times 100$$

Voltage Regulation

$$Regulation = \frac{|V_{nl}| - |V_l|}{|V_{nl}|}$$

Regulation = $$\frac{I_2 R_{02} \cos \phi \pm I_2 X_{02} \sin \phi}{{}_0V_2} \times 100$$

Losses in a Transformer

- Core loss (iron loss)
    - Hysterisis Loss
    - Eddy Current Loss
- Copper loss

Efficiency of Transformer

$$\text{efficiency} = \frac{\text{output}}{\text{input}}$$

$$\text{efficiency} = \frac{\text{output}}{\text{output + losses}} = \frac{\text{output}}{\text{output + Cu loss + iron loss}}$$

$$\eta = \frac{\text{input} - \text{losses}}{\text{input}} = 1 - \frac{\text{losses}}{\text{input}}$$

Condition for max. efficiency

$$\text{Cu loss} = I_1^2 R_{01} \quad \text{or} \quad I_2^2 R_{02} = W_{cu}$$

$$\text{Iron loss} = \text{hysteresis loss} + \text{eddy current loss} = W_h + W_e = W_i$$

Considering primary side,

$$\text{Primary input} = V_1 I_1 \cos \phi_1$$

$$\eta = \frac{V_1 I_1 \cos \phi_1 - \text{losses}}{V_1 I_1 \cos \phi_1} = \frac{V_1 I_1 \cos \phi_1 - I_1^2 R_{01} - W_i}{V_1 I_1 \cos \phi_1}$$

$$= 1 - \frac{I_i R_{01}}{V_1 \cos \phi_1} - \frac{W_i}{V_1 I_1^2 \cos \phi_1}$$

Differentiating both sides with respect to I_1, we get

$$\frac{d\eta}{dI_1} = 0 - \frac{R_{01}}{V_1 \cos \phi_1} + \frac{W_i}{V_1 I_1^2 \cos \phi_1}$$

For η to be maximum, $\frac{d\eta}{dI_1} = 0$. Hence, the above equation becomes

$$\frac{R_{01}}{V_1 \cos \phi_1} = \frac{W_i}{V_1 I_1^2 \cos \phi_1} \quad \text{or} \quad W_i = I_1^2 R_{0r} \quad \text{or} \quad I_2^2 R_{02}$$

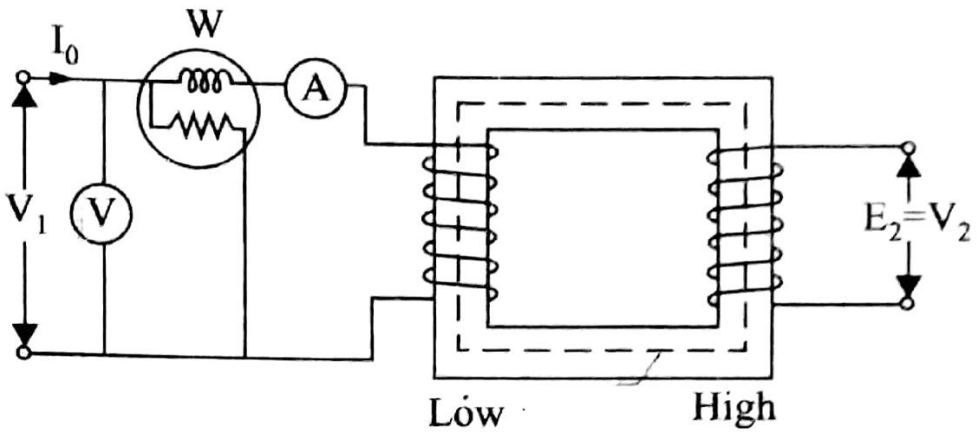
or

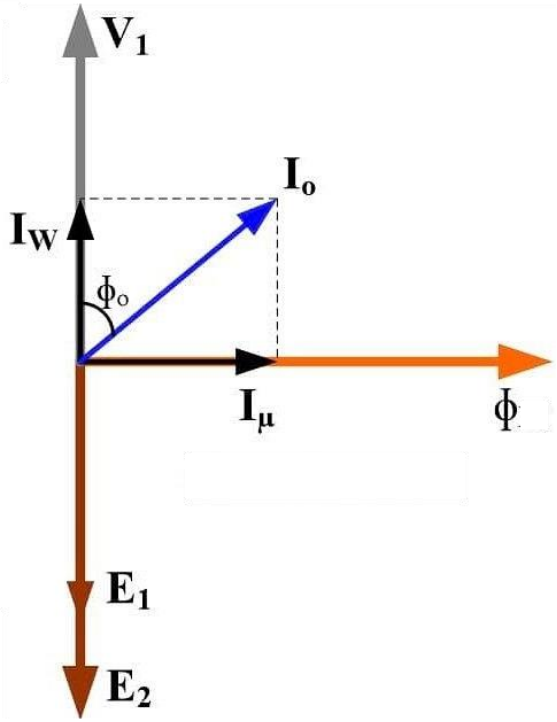
$$\text{Cu loss} = \text{Iron loss}$$

Testing of Transformer

- Open Circuit test (No load test)
- Short Circuit test

OC Test





- HV side kept as open
- Rated voltage is applied to LV side
- Cu loss is negligibly small
- Wattmeter reading gives core loss

$$W = V_1 I_0 \cos \phi_0 \quad \therefore \quad \cos \phi_0 = W/V_1 I_0$$

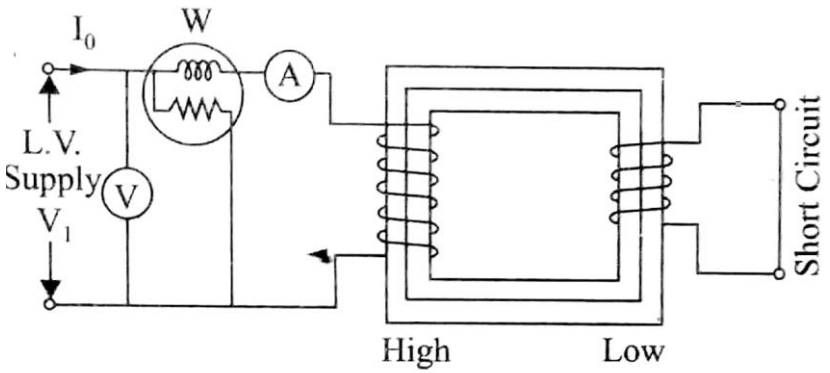
$\therefore$ $\quad I_\mu = I_0 \sin \phi_0, \; I_w = I_0 \cos \phi_0 \; \therefore \; X_0 = V_1/I_\mu$ and $R_0 = V_1/I_w$

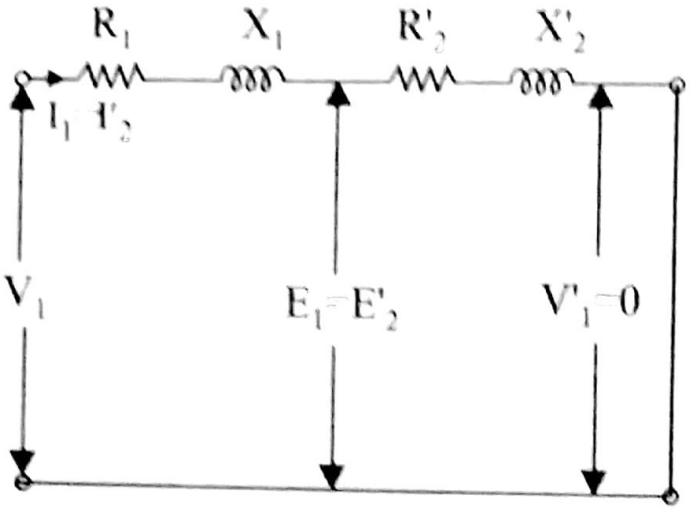
Or since the current is practically all-exciting current when a transformer is on no-load (*i.e.* $I_0 \cong$ and as the voltage drop in primary leakage impedance is small*, hence the exciting admittance Y_0 he transformer is given by $I_0 = V_1 Y_0$ or $Y_0 = I_0/V_1$

The exciting conductance G_0 is given by $W = V_1^2 G_0$ or $G_0 = W/V_1^2$

The exciting susceptance $B_0 = \sqrt{(Y_0^2 - G_0^2)}$

SC Test





- LV side kept as short circuited
- Rated current is applied to HV side
- Only 5 to 10 % of normal voltage is applied. Hence flux produced is very small
- Core loss is negligibly small
- Wattmeter reading gives copper loss

$$Z_{01} = V_{SC}/I_1$$

Also $W = I_1^2 R_{01}$

$\therefore$ $R_{01} = W/I_1$

$\therefore$ $X_{01} = \sqrt{\left(Z_{01}^2 - R_{01}^2\right)}$

